Numerator Layout

Scaler α

$$\frac{\partial \alpha}{\partial \vec{x}} = \begin{bmatrix} \frac{\partial \alpha}{\partial x_0} & \cdots & \frac{\partial \alpha}{\partial x_{n-1}} \end{bmatrix}$$

$1 \times n$ matrix

Exercise 1.

i) $$\frac{\partial(\vec{\alpha}^T \vec{x})}{\partial \vec{x}} = \vec{\alpha}^T$$

$(\vec{\alpha}^T \vec{x})$ is a scaler, since $\vec{\alpha}^T\vec{x}$ is an inner product, thus
$(1\times n \cdot n\times 1)$ $\vec{\alpha}^T$ is an $1\times n$ matrice & $\vec{x}$ is a $n\times 1$ matrice. $\Rightarrow 1\times 1$

$$(*) \quad \frac{\partial(\vec{\alpha}^T\vec{x})}{\partial \vec{x}} = \begin{bmatrix} \frac{\partial \vec{\alpha}^T\vec{x}}{\partial x_0} & \cdots & \frac{\partial \vec{\alpha}^T\vec{x}}{\partial x_{n-1}} \end{bmatrix}$$ where $\vec{\alpha}^T = [\alpha_0 \ldots \alpha_{n-1}]$

$$\vec{x} = [x_0 \ldots x_{n-1}]^T = \begin{bmatrix} x_0 \\ \vdots \\ x_{n-1} \end{bmatrix}$$

I can write this product as a sum

$\sum_{i=0}^{n-1} \alpha_i x_i$ thus $\frac{\partial}{\partial x_k}$ acts as a Kronecher delta δ_{ik}, being 0 when $k \neq i$ and 1 where $i = k$

$$\frac{\partial}{\partial x_k}\left(\sum_{i=0}^{n-1}\alpha_i x_i\right) = \frac{\partial \alpha_0 x_0}{\partial x_k} + \ldots + \frac{\partial \alpha_k x_k}{\partial x_k} + \ldots \frac{\partial \alpha_{n-1}x_{n-1}}{\partial x_k} = \alpha_0\delta_{0k} + \ldots \alpha_k \delta_{kk} + \ldots \alpha_{n-1}\delta_{n-1k}$$

$$= \alpha_k \quad \text{for } \forall k = 0 \ldots n-1$$

We can insert this into $(*)$ such that

$$\frac{\partial(\vec{\alpha}^T x)}{\partial x} = [\alpha_0 \ldots \alpha_k \ldots \alpha_{n-1}] = \vec{\alpha}^T$$

---

ii) $$\frac{\partial(\overset{1\times n}{a^T}\overset{n\times n}{A}\overset{n\times 1}{a})}{\partial a} = a^T(A + A^T)$$

$(a^T A a)$, with order $1\times n \cdot n\times n \cdot n\times 1$, is a scaler ($1\times 1$), thus

$$(**) \quad \frac{\partial(a^TAa)}{\partial a} = \overset{1\times n}{\begin{bmatrix} \frac{a^TAa}{\partial a_0} & \cdots & \frac{a^TAa}{\partial a_{n-1}} \end{bmatrix}}$$

$a^T = [a_0 \ldots a_{n-1}]$, $a = \begin{bmatrix} a_0 \\ \vdots \\ a_{n-1} \end{bmatrix}$

$A = \begin{bmatrix} \alpha_{00} & \cdots & \alpha_{0n-1} \\ & & \\ \alpha_{n-10} & & \alpha_{n-1n-1} \end{bmatrix}$

$$a^TAa = \sum_{i=0}^{n-1}\sum_{j=0}^{n-1} a_i \alpha_{ij} a_j$$

$$\frac{\partial(a^TAa)}{\partial a_k} = \frac{\partial}{\partial a_k}\left(\sum_{i=0}^{n-1}\sum_{j=0}^{n-1} a_i\alpha_{ij}a_j\right)$$

$$= \sum_{i=0}^{n-1}\sum_{j=0}^{n-1}\left(\frac{\partial a_i}{\partial a_k}\right)\alpha_{ij}a_j + \sum_{i=0}^{n-1}\sum_{j=0}^{n-1} a_i\alpha_{ij}\left(\frac{\partial a_j}{\partial k}\right)$$

Again $\frac{\partial}{\partial a_k}$ acts as δ_{ik} in the first term and δ_{jk} in the second term

$$= \sum_{j=0}^{n-1}\alpha_{kj}a_j + \underbrace{\sum_{i=0}^{n-1} a_i\alpha_{ik}}$$ component k of the inner product a^TA

$\sum_{j=0}^{n-1}\alpha_{kj}a_j$ must be an element of $(**)$, thus $\alpha_{kj}a_j$ is a scaler and equals its own transpose, thus we write

$= \sum_{j=0}^{n-1} a_j\alpha_{jk}$ = which is component k of a^TA^T, thus we see that summing over k components gives a $1\times n$ column matrice by the placing of the indices α_{jk}

and we get $$\sum_{k=0}^{n-1}\left(\sum_{i=0}^{n-1} a_i\alpha_{ik} + \sum_{j=0}^{n-1} a_j\alpha_{jk}\right) = a^TA + a^TA^T = a^T(A+A^T)$$

Exercise 1

(ii) $\frac{\partial (X-AS)^T(X-AS)}{\partial S} = -2(X-AS)^TA$

$nxm \cdot mx1$
$A \cdot S$

... $X \in \mathbb{R}^n$, $S \in \mathbb{R}^m$, $A \in \mathbb{R}^{n\times m}$ such that $AS \in \mathbb{R}^{n\times 1}$

$(X-AS)^T(X-AS)$, $(X-AS)$ can be written as $\underbrace{\sum_{i=0}^{n-1}(X_i - \sum_{j=0}^{m-1} A_{ij}S_j)}_{f(S)^2}$

Such that $(X-AS)^T(X-AS)$ can be written as

$$\frac{\partial}{\partial S_k} f(S)^2 = \frac{\partial}{\partial S_k}\sum_{i=0}^{n-1}(X_i - \sum_{j=0}^{m-1} A_{ij}S_j)^2 =$$

Chain Rule

$$= 2\sum_{i=0}^{n-1}(X_i - \sum_{j=0}^{m-1}A_{ij}S_j)\frac{\partial}{\partial S_k}\left(\sum_{j=0}^{m-1}A_{ij}S_j\right) = 2\sum_{i=0}^{n-1}(X_i - \sum_{j=0}^{m-1}A_{ij}S_j)A_{ik}$$

$$\frac{\partial}{\partial S_k} f(S)^2 = 2\underbrace{\sum_{i=0}^{n-1}(X_i - \sum_{j=0}^{m-1}A_{ij}S_j)A_{ik}}_{1\times 1} \quad \text{for } \forall\, k = 0,\dots, m-1$$

$(X-AS)^TA$ since the function $f(S)^2$ is a scalar

Thus $\frac{\partial (X-AS)^T(X-AS)}{\partial S} = -2(X-AS)^TA$

Second derivative with respect to S

$$-2\frac{\partial}{\partial S}(X-AS)^TA \;;\; -2\frac{\partial}{\partial S_k}\sum_{i=0}^{n-1}(X_i - \sum_{j=0}^{m-1}A_{ij}S_j)A_{ik}$$

$$= -2\sum_{i=0}^{n-1}\sum_{j=0}^{m-1}-\frac{\partial}{\partial S_k}(A_{ij}S_j)A_{ik} = 2\sum_{i=0}^{n-1}A_{ik}A_{ik} \;\forall\, k=0,\dots,m-1$$

Thus $\frac{\partial}{\partial S}(-2(X-AS)^TA) = 2A^TA$

$$\frac{\partial (Y-X\theta)^T(Y-X\theta)}{\partial S} = -2(Y-X\theta)^TX$$

$$\frac{\partial^2}{\partial S}(Y-X\theta)^T(Y-X\theta) = 2X^TX \quad \alpha H$$